School of Future Tech
Case Study Report

on

Online Expense Splitter System

by

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1. Introduction to the Case Study

In everyday situations such as group trips, shared meals, or collaborative activities, people often share expenses. Calculating how much each person should pay can become tedious and prone to errors when done manually.
This case study focuses on developing a **simple Online Expense Splitter system using C++** that calculates how much each participant should contribute to a shared expense. The system allows users to input the total expense, choose a currency type, and specify the number of participants. The program then automatically calculates the amount each person must pay.
The case study demonstrates how **basic programming concepts such as user input, conditional statements, switch cases, and arithmetic operations** can be used to build a practical and useful command-line application.

2. Problem Statement / Case Background (Abstract)

Background

When multiple people share expenses, determining each person's contribution can become confusing. Manual calculations may lead to incorrect results, especially when dealing with different currencies or larger groups.

Programming can simplify this process by automating the calculation and ensuring accuracy.

Abstract

This case study presents the design and implementation of an Online Expense Splitter system using the C++ programming language. The program takes user inputs such as the total expense amount, currency type, and number of participants. It then calculates the equal share for each participant.
The system also provides multiple currency options such as:

- INR (Indian Rupee)
- USD (US Dollar)
- EUR (Euro)
- GBP (British Pound)
- JPY (Japanese Yen)
- AUD (Australian Dollar)
- CAD (Canadian Dollar)

The program uses conditional logic and switch statements to process user choices and perform accurate calculations.

3. Case Study Design

The Online Expense Splitter system is designed as a **console-based application** with the following stages:

1. User Interaction

The program begins by displaying a menu with two options:

- Split Expense
- Exit Program

The user selects an option by entering a number.

2. Input Collection

If the user selects the split expense option, the system collects the following information:

- Total expense amount
- Currency type
- Number of participants

3. Currency Selection

The program uses a **switch-case structure** to determine the selected currency and assign the correct currency name.

4. Expense Calculation

The system calculates the share of each participant using the formula:

Total Expense ÷ Number of Participants

5. Result Display

Finally, the program displays how much each participant must pay in the selected currency.

4. Methods & Technologies Applied in the Case Study

Methods Used

1. User Input Handling
The program uses cin to collect data from the user such as:

- Expense amount
- Currency type
- Number of participants

2. Conditional Statements : Conditional statements (if, else if, else) are used to control the program flow based on user choices.

3. Switch Case Structure : A switch statement is used to assign the appropriate currency name depending on the user’s currency selection.

4. Arithmetic Calculation : The program calculates each person's share using a simple division formula.

5. Error Handling : Basic validation is included to ensure that the number of participants is greater than zero.

Technology Stack Used

- **Programming Language : C++**
- **Compiler : g++**
- **Development Environment: VS Code**

5. Case Study Implementation Details and Snapshots.

The implementation of the Online Expense Splitter consists of the following steps:

Step 1 : Display Menu
The program begins by showing a simple menu to the user.

Step 2: Collect Expense Details
The user enters:
- Total expense amount
- Currency type
- Number of participants

Step 3: Currency Selection Logic
The program determines the selected currency using a switch statement.

Step 4: Expense Calculation
The system calculates the share using:
share = totalExpense / participants;

Step 5: Display Output
The program displays the result to the user.

Snapshots:

```
● adityayadav@Aditya-Yadavs-MacBook-Air c++ % ./expense_splitter
  Online Expense Splitter
  1. Split Expense
  2. Exit
  Enter your choice: 1
  Enter total expense amount: 60007
  Select Currency Type
  1. INR (Indian Rupee)
  2. USD (US Dollar)
  3. EUR (Euro)
  4. GBP (British Pound)
  5. JPY (Japanese Yen)
  6. AUD (Australian Dollar)
  7. CAD (Canadian Dollar)
  Enter currency choice: 1
  Enter number of payers: 3

  Each Person Pays: 20002.3 INR
○ adityayadav@Aditya-Yadavs-MacBook-Air c++ %
```

6. Case Study Results and Conclusion.

Findings:

The Online Expense Splitter successfully demonstrates how simple programming techniques can automate everyday calculations. The program effectively performs expense splitting while supporting multiple currencies.

Key observations include:

- Accurate expense distribution among participants
- Easy-to-use menu interface
- Support for multiple currency types
- Prevention of invalid participant input

Conclusion:

This case study demonstrates the development of a **console-based expense splitting system using C++**. The program uses fundamental programming constructs such as conditional statements, switch cases, and arithmetic operations to solve a real-world problem.

The system provides a simple and efficient method for calculating shared expenses and can be extended further with additional features such as uneven expense distribution, file storage, or graphical user interfaces.

7. References

- C++ Programming Language Documentation
- Standard Template Library (STL) Documentation
- Online C++ tutorials and educational resources
- Programming textbooks on basic C++ concepts and control structures